



A solar cooker equipped with a bottom parabolic reflector and absorber plate integrated with organic PCM: Mathematical modelling, exergy and economic analyses

M.A. Tawfik^{a,*}, Atul A. Sagade^b, A.A. El-Sebaii^c, A.M. Khallaf^d, Abhishek Saxena^e, Hanan M. El-Shal^a, W.E. Abd Allah^a

^a Agricultural Engineering Department, Faculty of Agriculture, Zagazig University, 44511, Zagazig, Egypt

^b Departamento de Ingeniería Mecánica, Universidad de Tarapacá, Arica, Chile

^c Department of Physics, Faculty of Science, Tanta University, Tanta, Egypt

^d Department of Basic Science, Misr Higher Institute for Engineering and Technology, Mansoura, Egypt

^e Mechanical Engineering Department, School of Engineering, Dev Bhoomi Uttarakhand University, Dehradun, Uttarakhand, India

ARTICLE INFO

Keywords:

Solar cooker
Absorber plate with PCM
Mathematical model
Thermal performance parameters
Economic analysis
Exergy analysis

ABSTRACT

The present work proposes mathematical models for different configurations of box-type solar cookers (SBCs) with a tracking-type bottom parabolic reflector (TBPR) integrated with paraffin wax as a phase change material (PCM), aiming to enhance the thermal performance of the solar cookers. Analytical models for the proposed configurations of SBC without TBPR and PCM, SBC with TBPR + without PCM, SBC without TBPR+ with PCM, and SBC with TBPR and PCM, which are denoted as SC1, SC2, SC3, and SC4, respectively, are formulated and validated with the experimental results under different operating and climatic conditions. Exergy and economic analyses for the investigated configurations were conducted. It is found that there is a good agreement between the calculated and measured temperatures of the different elements in the investigated configurations, with a relative percentage difference (RPD) not more than 10 % for most cases. The average values of the adjusted (P) and standard (P_s) cooking powers, at a temperature difference of 50 °C, are 22.19 and 18.20 W, respectively. The utilization efficiency (η_{th}) of the SC4 solar cooker is higher than that of the SC1, SC2, and SC3 by approximately 32.62 %, 19.24 %, and 4.59 %, respectively. The exergy and economic analyses showed that the SC4 is more efficient than other configurations, with the lowest lifetime cost of cooking a meal of 0.021 \$.

1. Introduction

Globally, cooking is one of the most energy-consuming domestic activities, accounting for about 30 % of the total energy required for the residential sector in many developed countries [1], and it consumes a substantial amount of fossil fuels such as kerosene, LPG, natural gas, etc. [2]. Thus, household energy use for cooking is considered one of the main concerns in energy consumption. A series of challenges concerning energy demands alongside the security of water and food supply will confront societies around the world [3]. In the same context, the continuous increase in prices of cooking conventional fuels and electricity in rural and remote areas, as well as in urban and semi-urban areas, has become a global issue of the present era [4]. Therefore, harnessing and implementing renewable energy is one of the most viable

and greenest routes to mitigate the greenhouse gas emissions that contribute aggressively to global warming [5]. Thus, there is an urgent need for a sustainable, free, and green energy source for the cooking process with widespread availability and low prices, which is represented in solar energy, which is considered the most sustainable energy source for such intensive energy consumption activities like the cooking process [6]. Accordingly, solar energy is considered one of the most viable and reliable energy sources for different household processes such as drying, water heating, and cooking foodstuffs [7]. Solar cooking is deemed one of the most effective, reliable, and eco-friendly techniques in terms of retention, high nutrition for food value, and reducing drudgery [8,9]. Numerous attempts have been carried out to resolve the drawback of using SBC for just a short time daily. Among the different techniques of heat storage, latent heat storage is a thermal energy

* Corresponding author.

E-mail address: mohamed.ezeldien@yahoo.com (M.A. Tawfik).

<https://doi.org/10.1016/j.energy.2026.140064>

Received 23 June 2025; Received in revised form 21 December 2025; Accepted 14 January 2026

Available online 15 January 2026

0360-5442/© 2026 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

storage (TES) employing a phase change material (PCM) that stores a large amount of heat during the phase change from solid to liquid [10–12].

Several attempts to improve the thermal performance of the solar cooker using PCM were presented in the literature; a few of them are the modified cooker incorporated with Acetanilide [13], Stearic acid [14], Erythritol [15], and cooking vessels charged with Erythritol and sunflower oil [16]. Coccia et al. [17] studied the thermodynamic properties of 2.5 kg of commercial-grade xylitol as PCM through practical experiments on a box-type solar cooker. Results show that the average cooling time of the load was about 148 % (in the range of 110–80 °C) longer compared to the same cooker without PCM. Osei et al. [18] incorporated erythritol (2.5 kg) as PCM into a solar-powered electric box cooker. The obtained results of the modified cooker showed that the efficiency is about 77 %. Cuce et al. [19] tested a double-glazed solar cooker integrated with 8 k beeswax beneath the absorber plate. The solar cooker demonstrated high capability to maintain water temperature at 40 °C for long hours in the evening, as well as F1 was about 0.08, and the cooker's thermal efficiency is found to be about 7.47 %.

Palanikumar et al. [20] successfully developed a box-type solar cooker integrated with two PCMs: the mixture of waste cooking oil and $C_4H_4O_3$ and a nanocomposite PCM, where they found that the bar plate absorber coated with $MgAl_2O_4$ /Ni-doped, Fe_2O_3 nanoparticles, and integrated with PCM based on the nanocomposite increases the internal temperature of the cooker to 164.12 °C, and bar plate absorber up to 163.74 °C with the cooker's highest thermal efficiency of 56.21 %.

Kumar et al. [21] developed a double glass cover box-type solar cooker by adding 144 aluminum capsules filled with organic PCM (RT64HC) that were placed horizontally over the absorber plate. Results show an explicit improvement in the thermal performance of the cooker by achieving the highest thermal efficiency (52.2 %) and the lowest overall heat loss coefficient (4.11 W/m²°C) compared to the same cooker without PCM capsules and the cooker with hollow capsules. Paraffin wax is one of the most attractive organic PCMs due to its preferences for odorless, softness, colorless, low melting temperature, and high boiling temperature, as well as its volumetric expansion during melting, making it suitable for a wide range of thermal applications [22–24]. This can reduce the cost of cooking a meal to 0.024 ± 0.0005 \$/meal [25].

It is known that the experimental work conducted to evaluate the thermal performance of any solar cooker incorporated into TES using PCM is always confined to certain operational conditions. Therefore, the theoretical analysis (e.g., numerical analysis, simulation models, etc.) is an essential tool to obtain deep insights into the thermal behavior of the solar cooker incorporated with a PCM as a heat storage medium under different conditions. From the available literature, the thermal performance of a double-exposure SBC integrated with this unique design of TES absorber plate packed with paraffin wax as a latent heat storage material hasn't been analyzed analytically till now.

Thus, previous works did not describe the detailed thermal model of this cooker integrated with thermal storage [26]; however, they helped to identify the potential key indicators, such as the maximum temperature attained by the test load, absorber plate, and PCM, along with a fair idea about the optical efficiency and heat loss characteristics, and helped in the reasonable rating and standardization of the cooker. This has allowed the manufacturers, designers, and end users to select the best option among the available cooker designs.

The literature review also depicted that solar cookers integrated with thermal storage suffered from their realistic rating. Most of the time, they are evaluated with the TPPs and corresponding test procedures, such as figures of merit (F1 and F2) [27,28], new figures of merit ($F1_{new}$ and $F2_{new}$) [29], and many times standardized cooking power [30,31], which do not comment on or accommodate the effect of thermal storage added/integrated in the solar box-type cookers. Because the addition/integration of thermal storage is expected to change the heat transfer and loss characterization of the solar cookers, these TPPs and corresponding test procedures may not provide realistic information on the

cooker's performance, rating, and standardization with thermal storage [32].

On the other hand, it is also beneficial to understand the detailed thermal behaviour of the cooker for developing improved future designs. Also, mathematical thermal models presented in the literature apply to a particular design of solar cookers integrated with thermal storage.

Thus, the novelty of the present work is to i) propose a mathematical model for SBC-TBPR integrated with TES-absorber plate, which provides a handy tool for enhancing future design interventions and predicting its thermal performance under different operating and climatic conditions, ii) perform an analytical analysis to compare four configurations of the cooker: SBC without TBPR and PCM (SC1), SBC with TBPR only (SC2), SBC with PCM only (SC3), and SBC with TBPR and PCM (SC4), and iii) conduct an economic and exergy analyses to compare the performance of the mentioned configurations.

2. Materials and methods

2.1. The design of solar cooker

As seen in Fig. 1a, the used cooker features a double-walled wooden cooking cabinet with a 3 mm thickness inclined single-sheet glass cover at 30° on the horizontal plane. This cabinet has a transparent bottom covered by a double glass cover of 3 mm thickness each, separated by an air gap of 5 mm. The internal sides and back wall of the cooking cabinet with loading door were covered with 3 mm in thickness glass mirrors.

The tracking-type bottom parabolic reflector (TBPR) is a wooden parabolic-shaped reflector covered by an anodized aluminium sheet with reflectivity of 85 %, which was placed below the cooking cabinet. In order to operate the cooker as a double exposure cooker, TBPR is carried on six wooden sills, which are attached to six trestles below the cooking cabinet to act as a booster reflector to intensify more radiation onto the bottom of the TES-absorber plate. The TES-absorber is a unique double-walled absorber plate (made of copper) with two indented dips, filled with paraffin wax as PCM, as depicted in Fig. 1b.

Due to the good results obtained previously from the experiments in that investigation for evening cooking purposes according to Ref. [26], a mathematical model is established in the current study and employed to assess the thermal performance of this novel design of the absorber plate with PCM under different conditions.

2.2. Experiments and procedures

The water heating tests (by increasing water temperature from ambient temperature to 95 °C) were carried out at Zagazig, Egypt (30.58°N, 31.50°E) on clear sunny days in the period from 15 December to 30 September to evaluate the thermal performance of the present design of solar cooker. Generally, the weather conditions during seasons at the experiment location were: average yearly ambient temperatures ranging between 20 to 40 °C, and solar radiation more than 700 W/m². Water heating tests were performed at ± 90 min of the local solar noon under the environmental conditions: $I_T \geq 700$ W/m²; $20^\circ\text{C} \leq T_a \leq 40^\circ\text{C}$, and wind speed (V) ≤ 1.5 m/s, where I_T is the total solar radiation in the direct beam direction, and T_a is the ambient temperature. The temperatures of the solar cooker's components were measured using two digital thermometers, whereas T_a and V were measured using a temperature-humidity meter and an air velocity meter, respectively. I_T was measured using a solar power meter with the help of a sun tracking guide bolt mounted on an inclined plate, as shown in Fig. 1a and b.

2.3. Mathematical model

Mathematical modeling for the solar cooker with and without TBPR, as well as the solar cooker with and without TBPR and PCM, has been carried out in previous works of Tawfik et al. [26,33], respectively. In

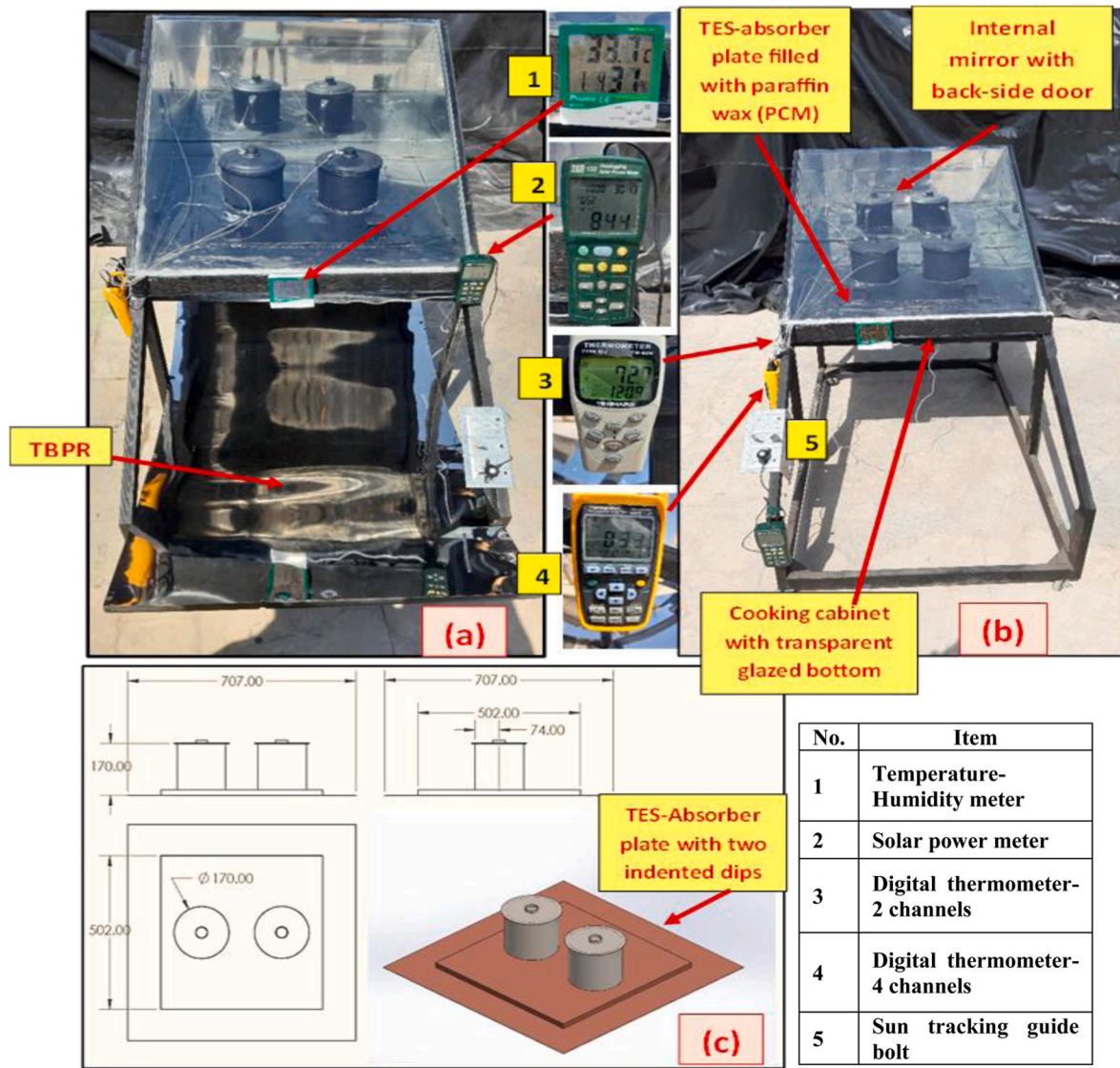


Fig. 1. The solar cooker based on Ref. [26]: solar cooker equipped with TBPR in case of double-exposure mode; b) solar cooker without TBPR; c) schematic diagram of the TES-absorber plate.

the present study, the effect of the PCM on the thermal performance of the solar cooker with TBPR has been investigated. Comparisons between the thermal performance of the four systems of the solar cooker, i.e., without TBPR (SC1), solar cooker with TBPR (SC2), solar cooker with PCM only (SC3), and solar cooker with TBPR and PCM (SC4), have been carried out using computer programs formulated in Pascal language. Energy balance equations for the various components of solar cookers, including the absorber plate, back plate, enclosed air, cooking fluid (water), cooking vessel, TBPR, and PCM, are written using the various heat transfers for the solar cooker with TBPR and PCM shown in Fig. 2.

The energy balance equations are written under the assumptions of: (i) the heat capacities of the different parts of the cooker are very small compared to that of the cooking fluid, so, they are neglected, (ii) the thickness of the cooking fluid is too small, therefore, there is no temperature gradient throw it, (iii) There is no air leakage from the cooker's interior. Using the above assumptions, the energy-balance equations for different parts of the cooker with and without PCM can be written as follows:

(a) For the charging mode

For the glass cover

$$I\alpha_g A_g + h_{tpg}(A_p - nA_t)(T_p - T_g) + h_{cag}A_g(T_{air} - T_g) = h_{cga}A_g(T_g - T_a) + h_{rgs}A_g(T_g - T_s), \quad (1)$$

For the enclosed air

$$h_{cpa}(A_p - nA_b)(T_p - T_{air}) + h_{cva}n(A_{lv} + A_t)(T_v - T_{air}) = h_{cag}A_g(T_{air} - T_g) + U_s A_s(T_{air} - T_a), \quad (2)$$

For the cooking vessel

$$I\tau_g \alpha_v n A_t + \left(\frac{K_p}{X_p}\right) n A_{bottom}(T_p - T_v) = h_{cva} n A_{lv}(T_v - T_w) + h_{cva} n(A_{lv} + A_t)(T_v - T_{air}), \quad (3)$$

For the absorber plate

$$I\tau_g \alpha_p(A_p - nA_t) = \left(\frac{K_p}{X_p}\right) A_p(T_p - T_{pcm}) + h_{tpg}(A_p - nA_t)(T_p - T_g) + \left(\frac{K_p}{X_p}\right) n A_{bottom}(T_p - T_v), \quad (4)$$

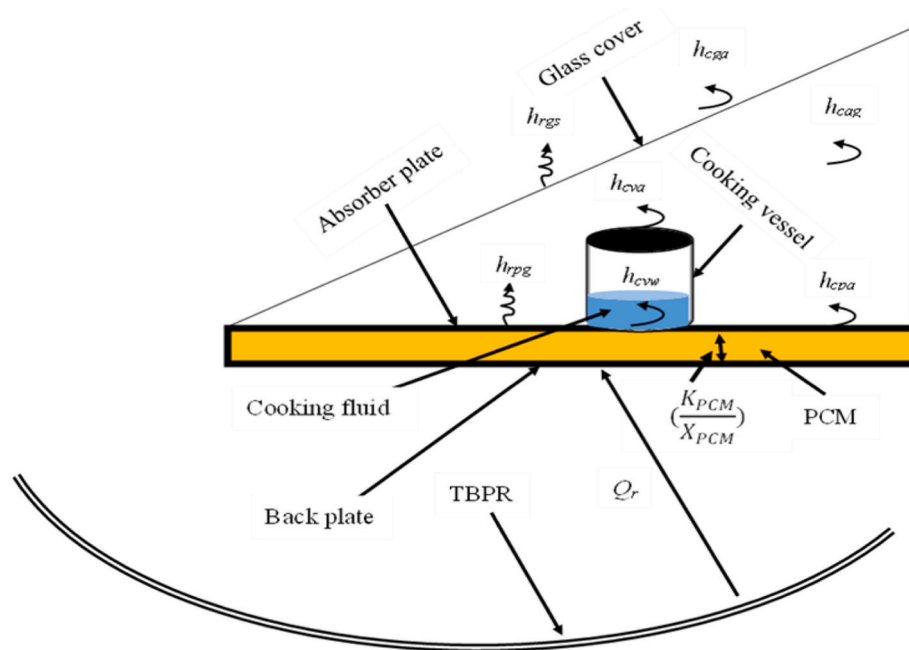


Fig. 2. The heat exchanges of the energy balance in the solar cooker.

For the back plate (the lower copper sheet of absorber plate)

$$Q_r = \left(\frac{K_p}{X_p}\right) A_b (T_b - T_{pcm}) + U_b A_b (T_b - T_a), \quad (5)$$

Where Q_r is the heat reflected from the TBPR, it can be calculated as:

$$Q_r = \eta_{opt} I A_b, \quad (6)$$

Where η_{opt} is the optical efficiency of the concentrator, and it depends on its optical properties [34]. h_{cW} is the convective heat transfer coefficient between the cooking vessel and water (the cooking fluid), h_{cva} is the convective heat transfer coefficient between the cooking vessel and the enclosed air. h_{cv} and h_{cva} are calculated using the following equations, as given by Holman et al. [35], Le Feuvre et al. [36], and Tiwari et al. [37].

$$h_{c,w} = \frac{0.686k}{L}(Gr.Pr)^{0.25} \left[\frac{Pr}{(1 + 1.05 Pr)} \right]^{0.25} + \frac{0.54k}{0.9d}(Gr.Pr)^{0.25} \quad (7)$$

$$h_{cva} = 0.54 \left(\frac{k_{air}}{X_{air}} \right) (Gr \cdot Pr)^{0.25} \quad (8)$$

where k_{air} is the thermal conductivity of the cooking fluid, L is the characteristic length, and is the diameter of the cooking vessel. h_{cga} and h_{rgs} are the convective and radiative heat transfer coefficients from the glass cover to the ambient air and sky, respectively. They can be calculated by the following correlations [38].

$$h_{cga} = 5.7 + 3.8V \text{ for } 0 \leq V \leq 5 \text{ m/s}, \quad (9)$$

$$h_{\text{rgs}} = \varepsilon_c \sigma (T_g^2 + T_s^2) (T_g + T_s). \quad (10)$$

Where h_{cag} is the convective heat transfer coefficient between the enclosed air and the glass cover. h_{cag} is calculated by the following equation [38].

$$h_{cag} = 0.54 \left(\frac{k_g}{X_a} \right) (Gr.Pr)^{0.25}. \quad (11)$$

From Eqs. (1)–(5), temperatures of the glass cover (T_g), enclosed air (T_{air}), cooking vessel (T_v), absorber plate (T_p), and the back plate (T_b) are

obtained as:

$$T_g = \frac{I\alpha_g A_g + h_{tpg}(A_p - nA_t)T_p + h_{cag}A_g T_{air} + h_{cga}A_g T_a + h_{rgs}A_g T_s}{h_{tpg}(A_p - nA_t) + h_{cag}A_g + h_{cga}A_g + h_{rgs}A_g}, \quad (12)$$

$$T_{air} = \frac{h_{cpa}(A_p - nA_b)T_p + h_{cva}n(A_{lv} + A_t)T_v + h_{cag}A_gT_g + U_sA_sT_a}{h_{cpa}(A_p - nA_b) + h_{cva}n(A_{lv} + A_t) + h_{cag}A_g + U_sA_s}, \quad (13)$$

$$T_v = \frac{I\tau_g \alpha_p n A_t + \left(\frac{K_p}{\lambda_p}\right) n A_{bottom} T_p + h_{cvw} n A_{vw} T_w + h_{cva} n (A_{lv} + A_t) T_{air}}{\left(\frac{K_p}{\lambda_p}\right) n A_{bottom} + h_{cvw} n A_{vw} + h_{cva} n (A_{lv} + A_t)}, \quad (14)$$

$$T_p = \frac{I\tau_g\alpha_p(A_p - nA_t) + \left(\frac{K_p}{X_p}\right)A_pT_{pcm} + h_{TPG}(A_p - nA_t)T_g + \left(\frac{K_p}{X_p}\right)nA_{bottom}T_v}{\left(\frac{K_p}{X_p}\right)A_p + h_{TPG}(A_p - nA_t) + \left(\frac{K_p}{X_p}\right)nA_{bottom}}, \quad (15)$$

$$T_b = \frac{Q_r + \left(\frac{K_p}{X_p}\right) A_b T_{pcm} + U_b A_b T_a}{\left(\frac{K_p}{X_p}\right) A_b + U_b A_b}, \quad (16)$$

For the cooking fluid (water)

$$h_{cwf} n A_{vw} (T_v - T_w) = m_w C_w \frac{dT_w}{dt}, \quad (17)$$

For the phase change material (PCM)

$$\left(\frac{K_p}{X_p}\right)A_b(T_p - T_{pcm}) + \left(\frac{K_p}{X_p}\right)A_p(T_p - T_{pcm}) = \frac{M_{equ}}{A_p} \frac{dT_{pcm}}{dt}, \quad (18)$$

Where:

$$M_{equ} = m_{pcm} C_{s,pcm} \quad T_{pcm} < T_m, \quad (18a)$$

$$M_{\text{equ}} = m_{\text{pcm}} L_{\text{pcm}} \quad T_{\text{pcm}} = T_m + \delta, \quad (18b)$$

$$M_{\text{equ}} = m_{\text{pcm}} C_{l, \text{pcm}} \quad T_{\text{pcm}} > T_m \quad (18\text{c})$$

By substituting T_g , T_{air} , T_v , T_p and T_b on Eqs. (17) and (18), the solutions of the two equations can be written as:

$$M_1 \frac{dT_w}{dt} + a_1 = \overline{f_1(t)}, \quad (19)$$

$$\text{and } M_2 \frac{dT_{pcm}}{dt} + a_2 = \overline{f_2(t)}, \quad (20)$$

The values of M_1 , a_1 and $\overline{f_1(t)}$ are listed in Appendix A, also, the values of M_2 , a_2 and $\overline{f_2(t)}$ are listed in Appendix B.

(b) For the discharging mode

Using the previous assumptions given in section (a), the mathematical model for the solar cooker with TBPR and PCM under the discharging mode has been developed. The energy balance equations for the glass cover, the enclosed air, and the cooking vessel are the same as those under the charging mode, as given in Eqs. (1)–(3), respectively.

For the absorber plate

$$\left(\frac{K_p}{X_p}\right) A_p (T_{pcm} - T_p) = h_{tpg} (A_p - nA_t) (T_p - T_g) + \left(\frac{K_p}{X_p}\right) nA_{bottom} (T_p - T_v), \quad (21)$$

For the back plate

$$Q_r + \left(\frac{K_p}{X_p}\right) A_b (T_{pcm} - T_b) = U_b A_{back} (T_b - T_a), \quad (22)$$

Using Eqs. (23) and (24), temperatures of the absorber plate and the back plate are calculated as:

$$T_p = \frac{\left(\frac{K_p}{X_p}\right) A_p T_{pcm} + h_{tpg} (A_p - nA_t) T_g + \left(\frac{K_p}{X_p}\right) nA_{bottom} T_v}{\left(\frac{K_p}{X_p}\right) A_p + h_{tpg} (A_p - nA_t) + \left(\frac{K_p}{X_p}\right) nA_{bottom}}, \quad (23)$$

$$T_b = \frac{Q_r + \left(\frac{K_p}{X_p}\right) A_b T_{pcm} + U_b A_{back} T_a}{\left(\frac{K_p}{X_p}\right) A_b + U_b A_{back}}, \quad (24)$$

For the cooking fluid (water)

$$h_{cww} nA_{vw} (T_v - T_w) = m_w C_w \frac{dT_w}{dt}, \quad (25)$$

For the PCM (Paraffin wax)

$$\frac{M_{equ}}{A_p} \frac{dT_{pcm}}{dt} = \left(\frac{K_{pcm}}{X_{pcm}}\right) A_p (T_{pcm} - T_p) + \left(\frac{K_{pcm}}{X_{pcm}}\right) A_b (T_{pcm} - T_b), \quad (26)$$

For $T_{pcm} = T_m$

$$\frac{m_{pcm} L_{pcm}}{A_p \Delta t} = \left(\frac{K_{pcm}}{X_{pcm}}\right) A_{back} (T_{pcm} - T_b) + \left(\frac{K_{pcm}}{X_{pcm}}\right) A_p (T_{pcm} - T_p), \quad (27)$$

$$T_{pcm} = \left\{ \frac{m_{pcm} L_{pcm}}{A_p \Delta t} + \left(\frac{K_{pcm}}{X_{pcm}}\right) A_{back} T_b + \left(\frac{K_{pcm}}{X_{pcm}}\right) A_p T_p \right\} / \left[\left(\frac{K_{pcm}}{X_{pcm}}\right) (A_{back} + A_p) \right] \quad (28)$$

Solutions of Eqs. (25) and (26) are:

$$M_3 \frac{dT_w}{dt} + a_3 = \overline{f_3(t)}, \quad (29)$$

$$\text{and } M_4 \frac{dT_{pcm}}{dt} + a_4 = \overline{f_4(t)}, \quad (30)$$

The values of M_3 , a_3 and $\overline{f_3(t)}$ are listed in Appendix C, also, the values of M_4 , a_4 and $\overline{f_4(t)}$ are listed in Appendix D. The separation of

variables method was used for solving Eqs. (19), (20), (29) and (30) using the initial and boundary conditions that; at time $t = 0$, the temperatures of different parts of the system (T_g , T_{air} , T_v etc.) equal the ambient temperature (T_a). The solutions of Eqs. (19) and (20) for the system under the charging mode are written as:

$$T_w = \frac{\overline{f_1(t)}}{a_1} \left[1 - \exp\left(\frac{-a_1 t}{M_1}\right) \right] + T_{wi} \exp\left(\frac{-a_1 t}{M_1}\right) \quad (31)$$

$$T_{pcm} = \frac{\overline{f_2(t)}}{a_2} \left[1 - \exp\left(\frac{-a_2 t}{M_2}\right) \right] + T_{pwi} \exp\left(\frac{-a_2 t}{M_2}\right) \quad (32)$$

For the system under the discharging mode, solutions of Eqs. (29) and (30), are:

$$T_w = \frac{\overline{f_3(t)}}{a_3} \left[1 - \exp\left(\frac{-a_3 t}{M_3}\right) \right] + T_{wi} \exp\left(\frac{-a_3 t}{M_3}\right) \quad (33)$$

$$T_{pcm} = \frac{\overline{f_4(t)}}{a_4} \left[1 - \exp\left(\frac{-a_4 t}{M_4}\right) \right] + T_{pwi} \exp\left(\frac{-a_4 t}{M_4}\right) \quad (34)$$

The thermal resistance network (thermal energy flow) is presented in Fig. 3, noting that expressions for R1 to R12 are given in appendix E.

The utilization efficiency (η_u), specific (t_s), and characteristic (t_c) boiling times represent the most important evaluation parameters for the thermal performance of the solar cookers. η_u , t_s , and t_c are calculated using the following correlations [39,40].

$$\eta_u = \frac{(M_w C_w)(T_{w,max.} - T_a)}{\bar{I}_s A_p \Delta t}, \quad (35)$$

$$t_s = \frac{\Delta t A_c}{M_w} \quad (36)$$

$$t_c = \frac{t_s I_{ave}}{I_{ave}^*}. \quad (37)$$

Where Δt is the time required to achieve the maximum temperature of the cooking fluid, I_{ave} is the average value of the solar intensity (W/m^2) during the time interval of Δt , and I_{ave}^* is a reference value of the average solar intensity equal to $900 W/m^2$ [38,39].

The input parameters for the mathematical model of the four systems of the solar cooker are the climatic parameters (the ambient temperature, wind speed and solar radiation for experiments location, design parameters (the applicable values of the materials from which the solar cooker has been constructed) and operational parameters (the number of cooking vessels, mass and kind of the cooking fluid). It is assumed that, at the beginning of the experiments (time $t = 0$), temperatures of various components of the cooker equal the ambient air temperature. After a time interval Δt (2 min), these temperatures are calculated using the various heat transfer coefficients and different climatic parameters. The relevant parameters used in the mathematical calculations are presented in Table 1.

2.4. Exergy analysis

Exergy analysis describes the quality of energy engaged in the irreversible state represented in the maximum amount of useful work when it is moved from its initial state to an equilibrium state for any solar device. For the steady-flow state heating process during a finite time interval, the output exergy of the SC is gained by the heated fluid (i.e., water) kept within the cooking vessel as the temperature rises [41]. The input exergy (E_{xi}) is the exergy of the solar radiation incident onto the SBC with PCM, which can be calculated using the following equation presented by Ref. [42].

$$E_{xi} = \left[1 + \frac{1}{3} \left(\frac{T_a + 273}{T_{sun}} \right)^4 - \frac{4}{3} \left(\frac{T_a + 273}{T_{sun}} \right) \right] \bar{I} A_c \quad (38)$$

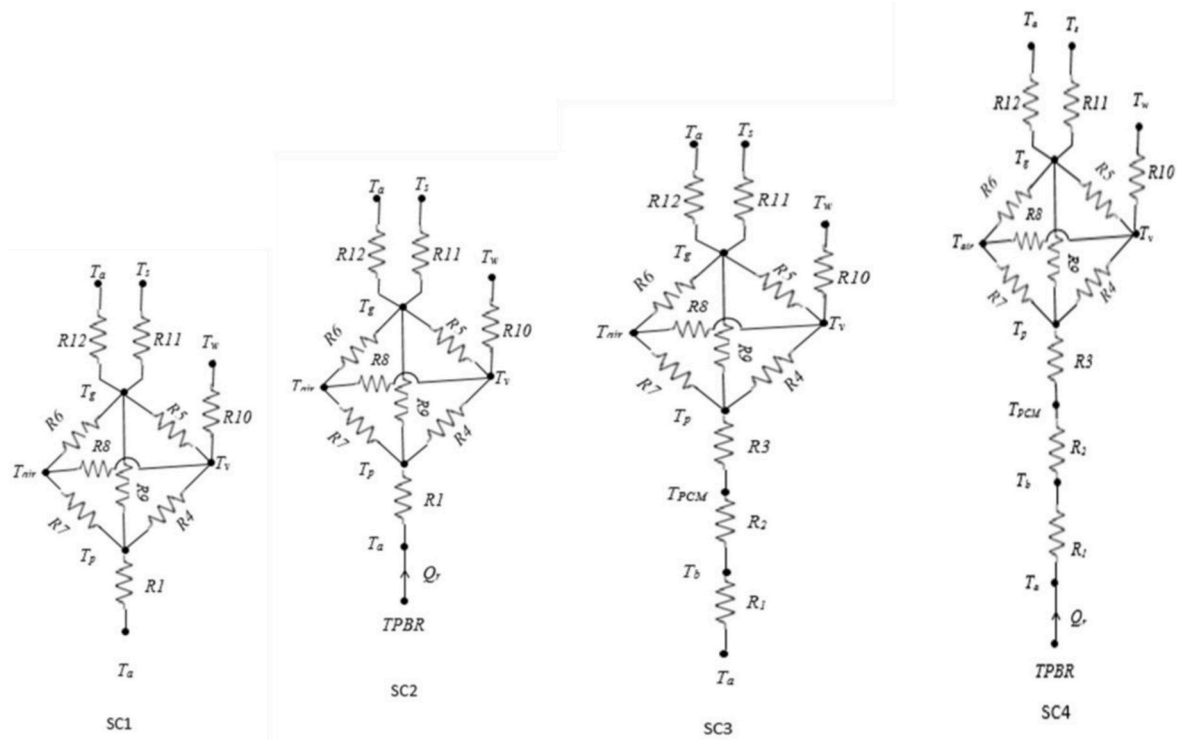


Fig. 3. The thermal resistance networks of the SBC without TBPR (SC1), SBC with TBPR (SC2), SBC with PCM only (SC3), and SBC with TBPR and PCM (SC4).

Table 1

Dimensions and thermophysical properties of the solar cooker's components.

Properties	Value
Material of cooking vessel	Aluminum
Material of absorber plate	Copper
Emissivity of the absorber plate ^a	0.95
Mass of cooking vessel included lid (kg/vessel)	0.576
Specific heat of cooking vessel (J/(kg K)) ^a	904
Specific heat of absorber plate (J/(kg K)) ^a	385
Specific heat of water (J/(kg K)) ^a	4186
Paraffin wax characteristics^b	
Melting start temperature (°C)	41
Melting onset temperature (°C)	60
Melting peak temperature (°C)	68
Melting offset temperature (°C)	88
Solid phase density (kg/m ³)	925
Liquid phase density (kg/m ³)	822
Thermal conductivity (W/m °C)	0.213
Solid phase specific heat (kJ/kg °C)	2.15
Liquid phase specific heat (kJ/kg °C)	2.15

^a Data from Tiwari et al. [37], Duffie and Beckman [38].

^b Data from Tawfik et al. [26].

where, $\bar{I}$ is the instantaneous irradiance (W/m²) recorded over a time interval Δt , and T_{sun} is the sun's surface temperature taken as ~ 5800 K [43].

Whereas, the average output thermal exergy (E_{xo}) of the solar cooker can be determined by Eq. (39) as given by Refs. [44,45].

$$E_{xo} = \frac{M_w \cdot C_w \left[(T_f - T_i) - \bar{T}_a \ln \frac{T_f}{T_i} \right]}{\Delta t} \quad (39)$$

where T_i and T_f are the initial and final temperature of water (°C), respectively, during a finite time interval, $\bar{T}_a$ is the average ambient temperature (°C) during the time interval of Δt (s).

Hence, the instantaneous exergy efficiency (η_x) is the ratio of exergy

gained by the fluid in the cooking vessel to the exergy of the incident solar radiation onto the solar cooker as:

$$\eta_x = \frac{E_{xo}}{E_{xi}} = \frac{M_w \cdot C_w \left[(T_f - T_i) - T_a \ln \frac{T_f}{T_i} \right]}{\left[1 + \frac{1}{3} \left(\frac{T_a + 273}{T_s} \right)^4 - \frac{4}{3} \left(\frac{T_a + 273}{T_s} \right) \right] \bar{I} A_p \Delta t} \quad (40)$$

Then, the daily exergy efficiency ($\eta_{x,daily}$) can be calculated in (%) as follows:

$$\eta_{x,daily} = \frac{\sum E_{xo}}{\sum E_{xi}} \times 100 \quad (41)$$

Where $\sum E_{xi}$ and $\sum E_{xo}$ are the summation of the input and output exergy throughout the cooking working time.

3. Results and discussions

The obtained results from the previous study of Tawfik et al. [33], of the mathematical modeling of the thermal performance of the solar cooker without TBPR (SC1) and with TBPR (SC2) solar cookers were compared to the modelled results derived from the experiments conducted by Tawfik et al. [26] of the thermal performance of two configurations of the same solar cooker without TBPR and PCM (SC3) and the solar cooker with TBPR and PCM (SC4), have been carried out.

3.1. The variation of experimental and theoretical temperatures of the different components of the solar cooker

Figs. 4 and 5 show the variations of calculated temperatures of the absorber plate (T_p), the back plate (T_b), cooking fluid (T_w), phase change material (T_{pcm}), cooking vessel (T_v), enclosed air (T_{air}), glass cover (T_g), ambient air (T_a), and the solar radiation intensity (I) for the SC3 and SC4 solar cookers, respectively. It is clear from Figs. 3 and 4 that the temperatures of all components of the two systems of solar cookers and also the solar radiation intensity increase with time until they achieve their

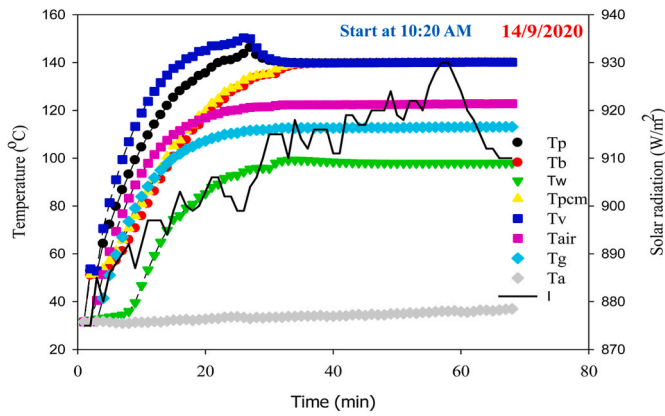


Fig. 4. Variations of different calculated temperatures of the absorber plate (T_p), back plate (T_b), cooking vessel (T_w), cooking fluid (T_v), enclosed air (T_{air}), glass cover (T_g), ambient air (T_a), and solar radiation (I) for the solar cooker with PCM only (SC3).

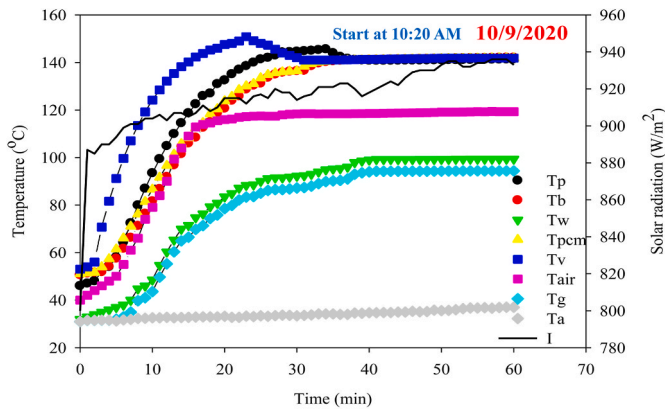


Fig. 5. Variations of different calculated temperatures of the absorber plate (T_p), back plate (T_b), cooking vessel (T_w), cooking fluid (T_v), enclosed air (T_{air}), glass cover (T_g), ambient air (T_a), and solar radiation (I) for the solar cooker with TBPR and PCM (SC4).

maximum values of 145.62 °C, 142.08 °C, 99.42 °C, 141.86 °C, 150.85 °C, 119.37 °C, 94.42 °C, 37.00 °C and 930.00 W/m² for T_p , T_b , T_w , T_{pcm} , T_v , T_{air} , T_g , T_a and I , respectively for the system SC3. The corresponding values for the system SC4 are 146.16 °C, 140.03 °C, 98.26 °C, 140.07 °C, 150.29 °C, 122.76 °C, 113.13 °C, 37.00 °C and 930.00 W/m², respectively.

To explore the thermal performance of the four systems of the solar cooker, numerical calculations are performed. To validate the theoretical models of the four SBC configurations, comparisons between experimental and theoretical results have been carried out. Figs. 6–11 illustrate the variations of measured and calculated temperatures of the absorber plate (T_p), cooking fluid (T_w), and phase change material (T_{pcm}) for the solar cookers, SC3 (Figs. 6–8) and SC4 (Figs. 9–11). It is seen from these figures that there is a good agreement between theoretical and experimental values. The relative percentage differences (RPD) between the average measured and calculated values of T_p , T_w , and T_{pcm} are found to be –2.98, 5.87 and 10.60, respectively, for the SC3 solar cooker and 3.49, 8.07 and 3.44 for the SC4 solar cooker.

For testing the accuracy of estimation of the temperatures of the absorber plate (T_p), water, as a cooking fluid (T_w), and phase change material (T_{pcm}), the mean bias error (MBE), the mean percentage error (MPE), and the root mean square error (RMSE) are calculated. These parameters are calculated by the following equations:

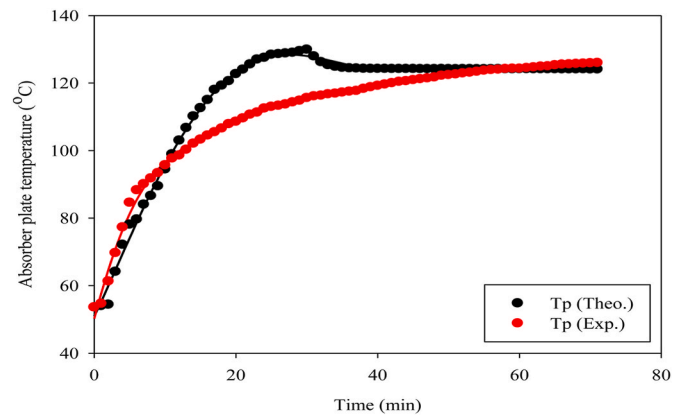


Fig. 6. Variations of the experimental and theoretical temperatures of the absorber plate (T_p) for the solar cooker with phase change material (PCM) (SC3).

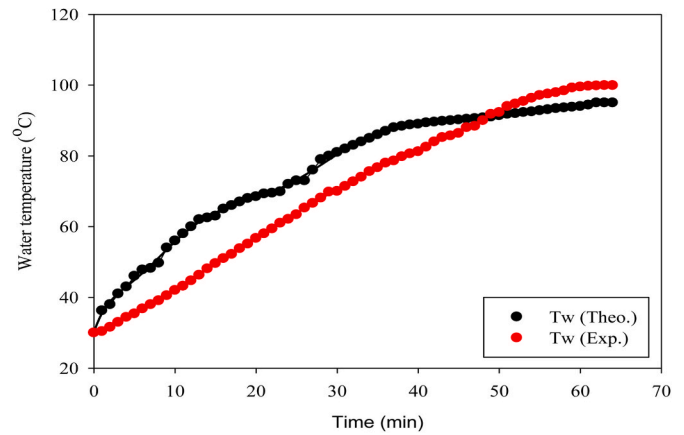


Fig. 7. Variations of the experimental and theoretical temperatures of the cooking fluid (water) (T_w) for the solar cooker with phase change material (PCM) (SC3).

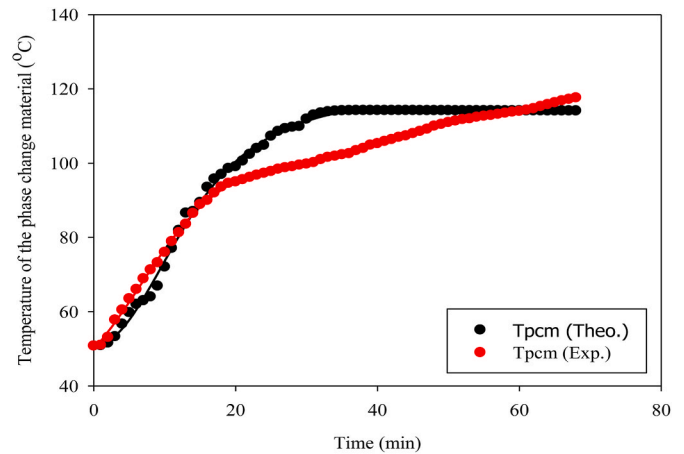


Fig. 8. Variations of the experimental and theoretical temperatures of the phase change material (T_{pcm}) for the solar cooker with phase change material (PCM) (SC3).

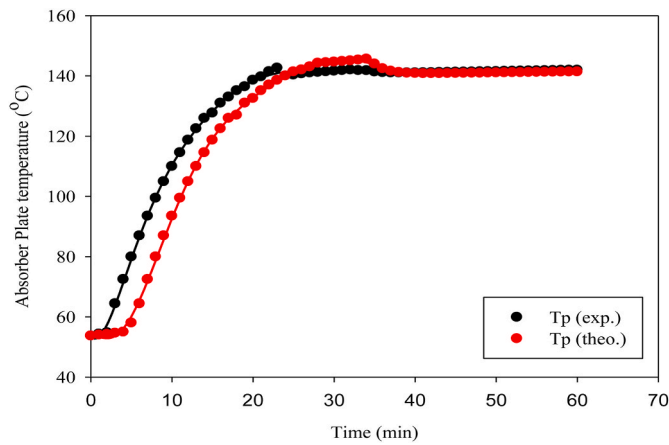


Fig. 9. Variations of the experimental and theoretical temperatures of the absorber plate (T_p) for the solar cooker with T racking-type bottom parabolic reflector (TBPR) and phase change material (PCM) (SC4).

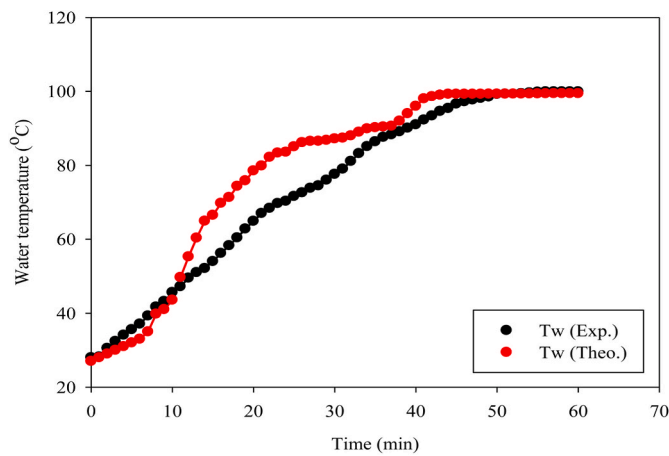


Fig. 10. Variations of the experimental and theoretical temperatures of the cooking fluid (water) (T_w) for the solar cooker with tubular bottom parabolic reflector (TBPR) and phase change material (PCM) (SC4).

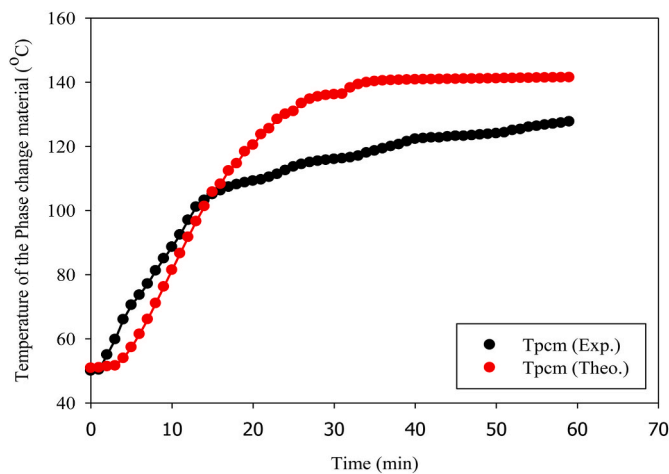


Fig. 11. Variations of the experimental and theoretical temperatures of the phase change material (T_{pcm}) for the solar cooker with tracking-type bottom parabolic reflector (TBPR) and phase change material (PCM) (SC4).

$$(MBE) = \frac{\sum (T_c - T_m)}{n}, \quad (42)$$

$$(MPE) = \frac{\sum \left(\frac{T_c - T_m}{T_m} \right) \times 100}{n}, \quad (43)$$

$$\text{and } (RMSE) = \left[\frac{\sum (T_c - T_m)^2}{n} \right]^{1/2}. \quad (44)$$

Where T_c and T_m are the calculated (from the mathematical model) and measured (from the experimental work) values of the temperature, n is the number of observations.

Table 2 presents the values of MBE, RMSE, and MPE of the temperatures of the absorber plate (T_p), water (T_w), and phase change material (T_{pcm}). It can be seen from **Table 2** that the values of MBE for the SC3 are found to be in the range of -4.27 to 10.60, and for the SC4 from 3.47 to 6.15. For the MPE, the values are 4.65–8.10 for the SC3 and 3.35–12.32 for the SC4. The corresponding values of the RMSE are 8.35–15.33 for the SC3 and 6.43–8.99 for the SC4. Most of these values represent an accepted range for these parameters, which verifies the accuracy of the proposed mathematical model. Therefore, the mathematical model can be used for predicting the thermal performance of different configurations of the solar cooker with an acceptable accuracy.

Using the mathematical models, comparisons between the thermal performance of the four configurations of solar cookers, i.e. SC1, SC2, SC3, and SC4, have been carried out. **Figs. 12–14** present the calculated values of temperatures of the absorber plate (T_p), cooking vessel (T_v), and the cooking fluid (T_w), respectively, for the solar cookers, SC1, SC2, SC3, and SC4. It is clear from **Figs. 12–14** that T_p , T_v , and T_w for the SC3 solar cooker are more than the SC1 and SC2 and slightly more than the SC4 solar cooker. For the absorber plate temperature (T_p), it is found that T_p of the SC3 is higher by 46.82 %, 27.06 %, and 1.15 % than those of the SC1, SC2, and SC4, respectively. The corresponding values for the temperatures of the cooking vessel (T_v) of the SC3 are higher by about 48.53 %, 27.97 %, and 1.87 %, and for (T_w) are 29.57 %, 13.52 %, and 0.63 % than those for SC1, SC2, and SC4, respectively.

3.2. The adjusted and standard cooking power

The adjusted (P) and standard (P_s) cooking powers represent the most important parameters that are used for evaluating the ability of the solar cooker for cooking foods [46].

They are calculated using the following equations according to Ref. [30].

$$P = M_f C_f \left[\frac{T_{fb} - T_{fa}}{t} \right], \quad (45)$$

$$P_s = \frac{P \cdot 700}{I_b} \quad (46)$$

Where I_b is the beam radiation during the time interval t , T_{fa} and T_{fb} are

Table 2

Values of RPD, MBE, MPE and RMSE for temperatures of the absorber plate (T_p), water (T_w) and phase change material (T_{pcm}) for the SC3 and SC4 solar cookers.

Configurations of the solar cooker	Accuracy parameters	T_p (°C)	T_w (°C)	T_{pcm} (°C)
TBPR and with PCM	RPD (%)	-2.98	5.87	10.60
	MBE (°C)	-4.27	4.91	10.57
	MPE (%)	4.65	6.10	8.10
	RMSE (°C)	8.35	7.46	15.33
With only PCM	RPD (%)	3.49	8.07	3.44
	MBE (°C)	4.03	6.15	3.47
	MPE (%)	3.35	12.32	3.13
	RMSE (°C)	7.72	8.99	6.43

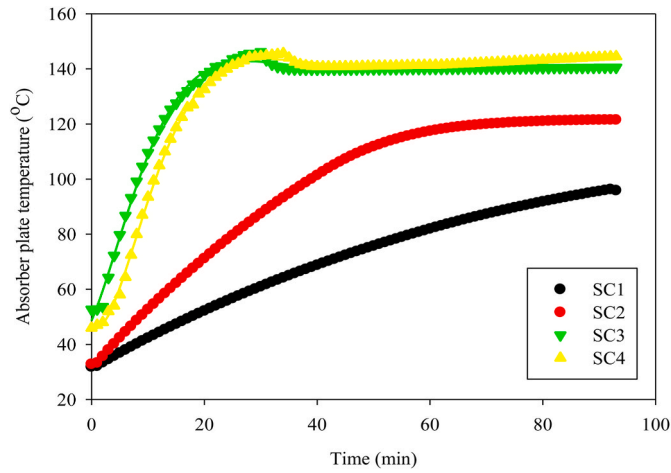


Fig. 12. Variations of the temperatures of the absorber plate (T_p) with time for the different four configurations of the solar cooker.

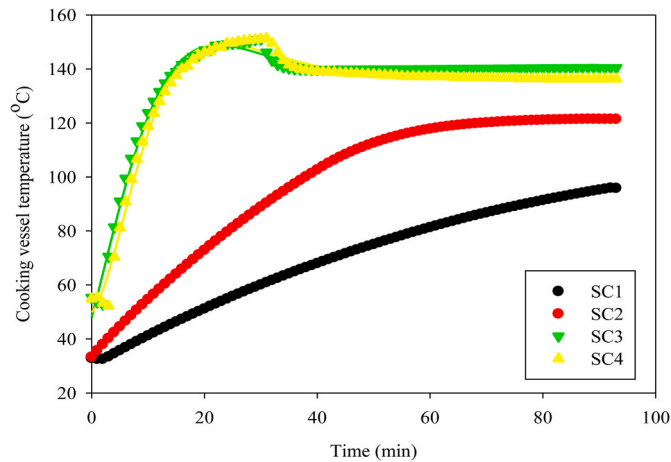


Fig. 13. Variations of the temperatures of the cooking vessel (T_v) with time for the different four configurations of the solar cooker.

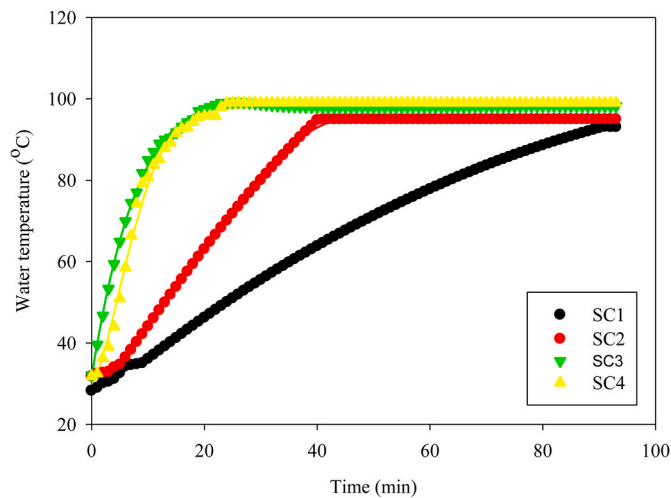


Fig. 14. Variations of the temperatures of cooking fluid (T_w) with time for the four different configurations of the solar cooker.

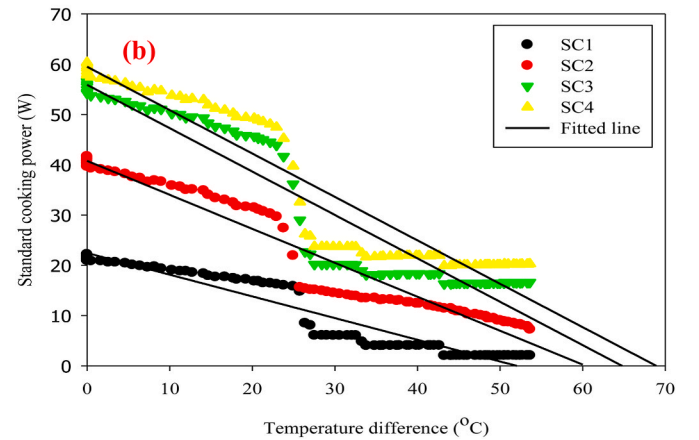
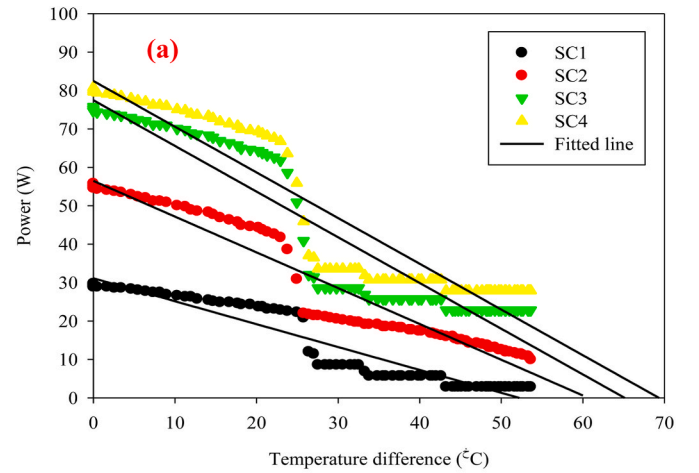


Fig. 15. Values of cooking power: a) the adjusted power, P ; b) the standard power, P_s , with different values of the temperature difference between the cooking fluid (water) and ambient air for the four different configurations of the solar cooker.

the initial and final instantaneous temperatures of the cooking fluid. Fig. 15a and b shows the variations of P and P_s for different values of temperature difference between the cooking fluid and the ambient air. Both P and P_s decrease with increasing temperature difference. The adjusted cooking power (P) as a function of the temperature difference between the cooking fluid and ambient air may be represented by the following equations:

$$P = 57.71 - 0.306 \Delta T, \text{ (For SC1)} \quad (47)$$

$$P = 67.71 - 0.914 \Delta T, \text{ (For SC2)} \quad (48)$$

$$P = 75.27 - 1.312 \Delta T, \text{ (For SC3)} \quad (49)$$

$$P = 80.27 - 1.312 \Delta T, \text{ (For SC4)} \quad (50)$$

The values of the regression coefficients (R^2) are 0.975, 0.978, 0.971, and 0.971 for the SC1, SC2, SC3, and SC4, respectively. The standard cooking power (P_s) is correlated with the temperature difference (ΔT) represented as:

$$P_s = 42.37 - 0.248 \Delta T, \text{ (For SC1)} \quad (51)$$

$$P_s = 60.42 - 0.859 \Delta T, \text{ (For SC2)} \quad (52)$$

$$P_s = 80.22 - 1.425 \Delta T, \text{ (For SC3)} \quad (53)$$

$$P_s = 88.89 - 1.445 \Delta T, \text{ (For SC4)} \quad (54)$$

The values of the regression coefficients (R^2) (Eqs. (51)–(54)), for the

standard power (P_s), are 0.976, 0.978, 0.972, and 0.972 for the SC1, SC2, SC3, and SC4, respectively. These results agree with those obtained by Refs. [38,39]. The values of the adjusted (P) and standard (P_s) cooking powers at a temperature difference of 50 °C are the standard values that govern the cooker performance, as suggested by Funk [30], for the international cooking power test procedures. The values of P for the SC1, SC2, SC3, and SC4, calculated by Eqs. (47)–(50) are obtained as 9.67, 14.67, 22.01, and 42.41 W, respectively. The corresponding values for P_s calculated by Eqs. (51)–(54) are 8.97, 16.64, 17.47, and 29.97 W for the SC1, SC2, SC3, and SC4, respectively. The average values of P and P_s , at a temperature difference of 50 °C, are 22.19 and 18.20 W, respectively. It is clear from the results plotted in Fig. 15a, b that the present solar cookers satisfy the standard procedure.

3.3. The utilization efficiency, specific and characteristic boiling times

The effect of the mass of the cooking fluid (water) on the utilization efficiency (η_u), specific, and characteristic boiling time has been investigated. Fig. 16 shows the effect of the cooking fluid mass (water) on the utilization efficiency (η_u). The efficiency increases with increasing water mass due to the increased heat capacity of water with increasing mass. Results indicated that the average η_u for the SC4 is about 61.60 %, which is more than those of the SC1, SC2, and SC3 by about 32.37 %, 19.25 %, and 4.61 %, respectively. The effect of mass of water on the specific (t_s) and characteristic (t_c) boiling times is presented in Fig. 17a and b. t_s and t_c decrease with increasing water mass. It can also be seen that t_s and t_c for the SC4 solar cooker are less than those for the SC1, SC2, and SC3. It is found from Fig. 17a that the average values of t_s for the SC4 (15.04 min m²/kg) are less than SC1, SC2, and SC3 by about 44.58 %, 31.73 %, and 7.95 %, respectively. On the other hand, average values of t_c for the SC4 (15.39 min m²/kg) are less than SC1, SC2, and SC3 by about 44.06 %, 31.26 %, and 7.84 %, respectively (see Fig. 17b). The values of t_c are obtained as 30.13, 25.13, 19.13, and 18.13 min m²/kg for SC1, SC2, SC3, and SC4 when boiling 1 kg of water, which is less than the value of 76.7 min m²/kg obtained by Ref. [47].

3.4. The daily efficiency of the solar cooker

The effect of mass of the cooking fluid (water) on the daily efficiency (η_d) is estimated. As seen from Fig. 18 that η_d increases as the water mass increases for the four configurations of the solar cookers, as expected. Also, the daily efficiency for the SC4 is more than that of SC1, SC2, and SC3. The daily efficiency for the SC1 varies from 5 % to 57 % as the values of water mass vary from 0.2 to 4.0 kg. The corresponding values for the SC2, SC3, and SC4 varied from 7 % to 66 %, from 9 % to 74 % and from 12 % to 86 %, respectively.

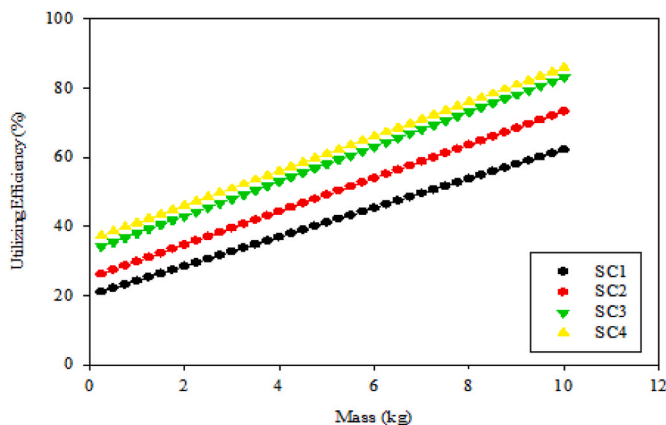


Fig. 16. Effect of the mass of the cooking fluid (water) on the total utilization efficiency (η_u) for the different configurations of the solar cooker.

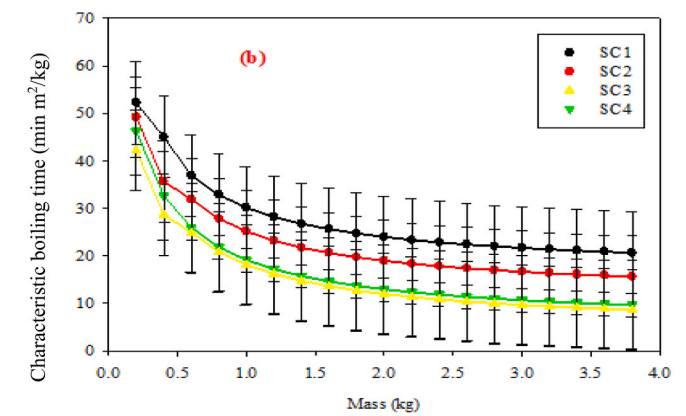
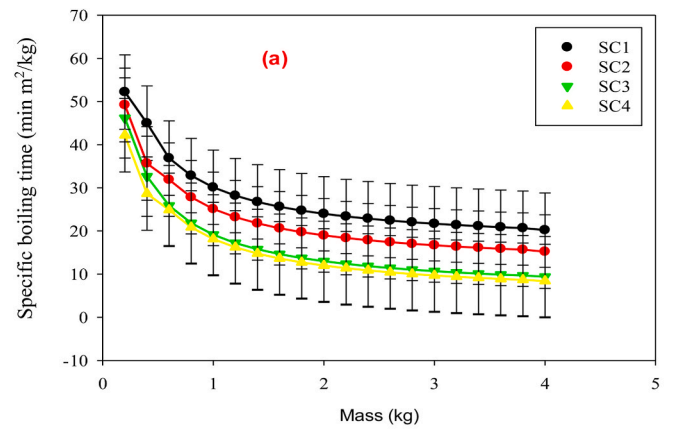


Fig. 17. Effect of the mass of the cooking fluid (water) on both: a) specific (t_s); b) characteristic (t_c) boiling time for the different configurations of the solar cooker.

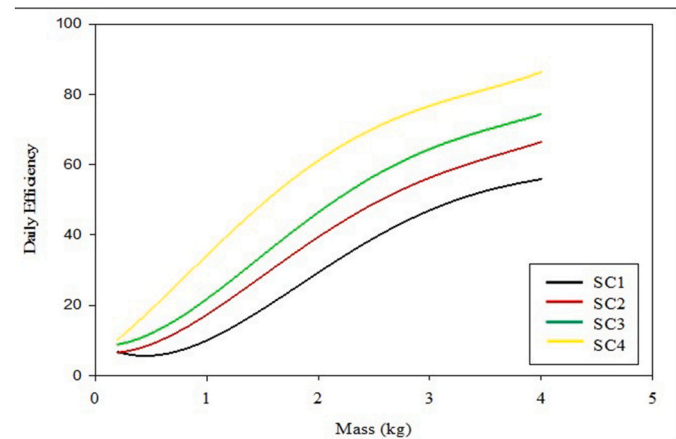


Fig. 18. Effect of the mass of the cooking fluid (water) on the daily efficiency (η_d) for the different configurations of the solar cooker.

3.5. The exergy analysis

The exergy efficiency is considered a crucial parameter that plays a vital role in describing the inefficiencies in any thermal system [48]. Fig. 19 a, b depicts the variation of the instantaneous exergy efficiency (η_{ex}) for different configurations of the SBC throughout the cooking period, alongside the daily exergy efficiency.

According to our previous work [33] for SC1 and SC2, Fig. 19a indicated that the instantaneous exergy efficiency of both cookers

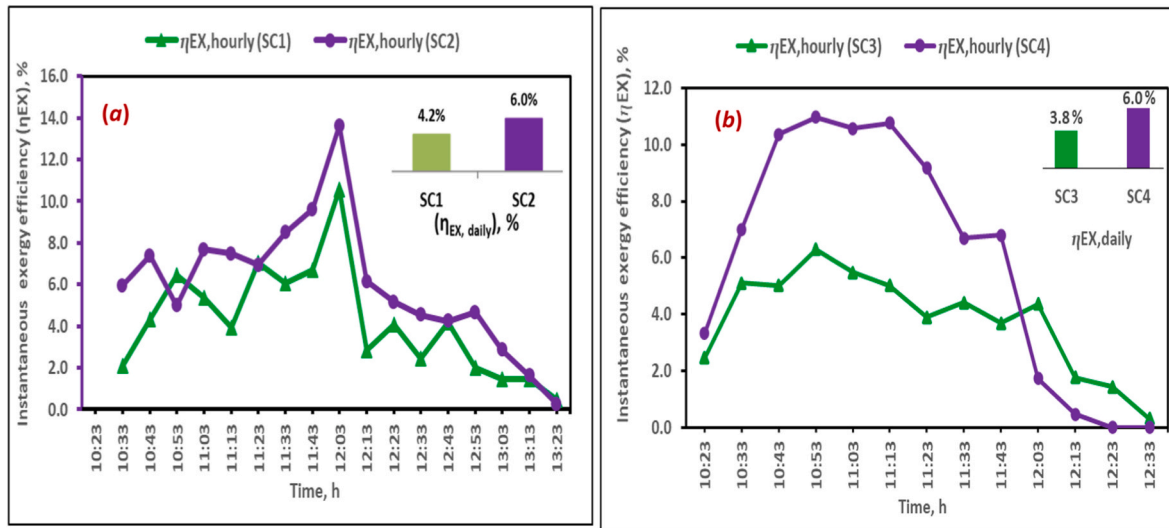


Fig. 19. The variation of instantaneous and daily exergy efficiency for the different configurations of solar cooker: a) SC1 and SC2 [32]; b) SC3 and SC4.

reached their peak values at solar noon with a clear increment in the instantaneous and daily exergy efficiency of SC2 (SC with TBPR) by about 22.05 % and 30 % over SC1 (SC without TBPR). This can be attributed to the crucial effect of the TBPR in reflecting more of the solar radiation towards the bottom of the absorber plate, which might lead to a high heating rate inside the cooking cabinet, which means it increases the net available thermal energy inside the cooker (net useful energy) after the irreversibility of heat loss through the upper and bottom glazed covers.

In the present work, the instantaneous and daily exergy efficiency are calculated for SC3 and SC4 according to the experimental results obtained by Ref. [33]. Fig. 19b reveals that the instantaneous exergy efficiency of both cookers reached their peak values before solar noon, which means earlier than SC1 and SC2, and continues to be stable during the period around the solar noon. It can be noticed that the instantaneous exergy efficiency values of SC3 and SC4 are more stable compared to SC1 and SC2, which suffer from continuous fluctuation throughout the cooking period, as seen in Fig. 19a and b.

It can be observed that there is one peak of the instantaneous exergy efficiency at solar noon of 13.9 % and 10.3 % for SC1 and SC2, respectively, with high fluctuation over the cooking period, while the

increment of the maximum instantaneous and daily exergy efficiency of SC4 was about 42.57 % and 36.66 %, respectively, over SC3.

Eventually, SC4 is considered the best configuration because of the high synergic effect of TBPR and TES-absorber plate in capturing and storing heat, which increases the net useful thermal energy inside the cooking cabinet by extending the heat delivery time and reducing the insufficiency of the cooker due to heat losses from the glazed covers.

3.6. The economic analysis

To conduct the economic analysis for this study, the lifetime cost for cooking a meal (LTCM) is estimated, which is mainly based on the total capital cost (C_C) and the number of annual cooked meals by the different configurations of solar cookers. The number of solar cooked meal annually (M_{lt}) were 600 and 1050 meal/year for SC1, SC2, respectively as given by Tawfik et al. [33], and 923 and 1126 for SC3 and SC4, respectively derived from Eq. (57) given by Tawfik et al. [26], when assuming 300 cooking days per year and cooking time of 4 h (240 min) per day (window of solar noon) for a typical year. Thus, M_{lt} can be calculated based on the reference time (τ_R) as Ref. [26]:

$$M_{lt} = \frac{\text{number of clear sun days } (n_s) \times \text{cooking time available in the solar noon } (CT)}{\tau_R} = \frac{300 \times 240}{\tau_R} \quad (55)$$

$$\tau_R = \frac{(m_f C_{pf})'_f \times COR}{A_p F \eta_o} \ln \left[\frac{\bar{G}_T \frac{T_{f,i} - T_a}{COR}}{\bar{G}_T \frac{T_{f,f} - T_a}{COR}} \right]$$

rest of the instantaneous exergy efficiency values are lower than those of SC3 and SC4, as seen in Fig. 19a and b. This may lead to degradation in the accumulated thermal energy inside the SC1 and SC2, consequently delaying the cooking process.

Regarding SC3 and SC4, it can be noticed that there are no peaks in the instantaneous exergy efficiency values, and these values seem to be more stable throughout the effective cooking period around the solar noon. This can be explained as the cooking vessels are placed in direct contact with the indented surface of the TES-absorber plate, the heat transfer from TES to the foodstuff being cooked will increase.

The obtained results showed that instantaneous and daily exergy efficiency values of SC4 were higher than those of SC3, wherein the

The calculations of economic analysis assume that the lifespan of the all configuration of solar cooker is 10 years, the discount rate (r) is 9.25 %, annual repair and maintenance cost (C_{OM}) varies up to 5 % of the capital cost [49], including the wage of manual sun-tracking and cleaning, which is considered as secondary work, and the salvage value (S) at the end of useful lifetime of cookers (n) can be assumed as 10 % of the net present cost (C_p), as reported by Ref. [50]. Accordingly, the net present cost (C_p) can be calculated as Ref. [51].

$$C_p = C_C + C_{OM} \left[\frac{(1+r)^n - 1}{r(1+r)^n} \right] - \left[\frac{S}{1+r^n} \right] \quad (56)$$

Table 3

The results of economic analysis.

Item	SC1 ^a	SC2 ^a	SC3	SC4
The whole wooden frame of the solar cooker with Trolley wheels, \$	45.90	45.90	45.90	45.90
Glass (Internal mirrors- upper and bottom glass cover), \$	34.5	34.5	34.5	34.5
Glass wool for insulation, \$	5	5	5	5
Labor wage (assembly cost), \$	20	20	20	20
TBPR (wooden chassis + reflective material), \$	–	30.6	–	30.6
TES-absorber plate (Material and manufacturing), \$	–	–	14	14
Price of local paraffin wax (PCM), \$	–	–	3	3
C _C , \$	105.4	136	122.4 ^a	153 ^a
C _{OM} , \$	23.52	30.35	25.70	32.13
Mlt, meal/year	600	1050	923 ^a	1126 ^a
C _A , \$/year	18.65	24.06	22.59	24.28
LTCM, \$/meal	0.031	0.023	0.024	0.021

^a Data derived from Tawfik et al. [26,33].

The C_{OM} can be determined by Eq. (57) as:

$$C_{OM} = f_{OM} \times C_C \quad (57)$$

Where f_{OM} is the operation and repair maintenance fraction (%), then the annual capital cost (C_A) can be calculated as:

$$C_A = \frac{C_P}{\left[\frac{(1+r)^n - 1}{r(1+r)^n} \right]} \quad (58)$$

Hence, the LTCM can be determined by the following equation:

$$LTCM = \frac{C_A}{M_{lt}} \quad (59)$$

The results of economic analysis for SC1, SC2, SC3, and SC4 are shown in Table 3. The capital cost (C_C), operation, maintenance cost (C_{OM}), and annualized costs (C_A) of SC2 and SC4 are higher than those of SC1 and SC3, due to the costs of material and manufacturing of the TBPR. SC4 has the highest C_C (153 \$), which is the costliest among all other configurations of the solar cooker due to the extra costs of TES-absorber plate, PCM, and TBPR.

However, results show that the LTCM of SC4 (0.021 \$) is lower than SC1, SC2, and SC3 by about 32.25 %, 8.70 % and 12.50 %, respectively. This can be attributed to the high number of cooked meals annually by SC4 (1126 meals/year) compared to the other configurations of the solar cooker. Hence, results elucidate the synergic effect of the TES-absorber plate with TBPR in increasing and maintaining a high amount of thermal energy inside the cooker, which led to shortening the cooking time daily, consequently increasing the annual number of meals cooked by SC4.

4. Conclusions

The conclusions of the current study can be summarized as follows:

- The mathematical model shows a good agreement between the measured and calculated, where the relative percentage difference (RPD) of the elements' temperatures of the SBC not exceeded 10 % for most cases.
- The mathematical model shows the most acceptable accuracy for SC4 values, shows the of MBE (3.47–6.15), MPE (3.35–12.32), and RMSE (6.43–8.99) compared to the other configurations of SBC.

- Among the different configurations of SBC, SC4 achieved the highest values of the adjusted (P) and standard (Ps) cooking powers of 42.41 W and 29.27 W, respectively, at a temperature difference of 50 °C.
- Results revealed that the average utilization efficiency (η_u) for the SC1, SC2, SC3, and SC4 were about 41.66, 49.74, 58.76, and 61.60 %, respectively. Thus, it can be found that SC4 has the highest difference in average utilization efficiency by about 32.37 % over SC1.
- The lowest average value of specific boiling time (t_s) of 15.04 m²/kg is achieved by SC4, which is less than SC1, SC2, and SC3 by about 44.58 %, 31.73 %, and 7.95 %, respectively.
- The average value of characteristic boiling time (t_c) of SC4 (15.39 min m²/kg) is less than SC1, SC2, and SC3 by about 44.06 %, 31.26 %, and 7.84 %, respectively.
- The daily efficiency (η_d) for all configurations of SBC decreases with increasing water mass. Results show that SC4 has the highest value of η_d varies from 12 % to 86 % as the water mass varies from 0.2 to 4.0 kg.
- SC3 and SC4 showed high and stable values of instantaneous exergy efficiency compared to SC1 and SC2 that suffer of severe fluctuation, particularly throughout the solar noon. Due to the synergic effect of TBPR and TES- absorber plate integrated with the SBC, an explicit increase in the instantaneous and daily exergy efficiency of SC4 by about 42.57 % and 36.66 %, respectively, over SC3.
- Due to the high number of annual cooked meals of SC4 (1125 meal/year), SC4 has the lowest lifetime cost of cooking a meal (LTCM) of 0.021 \$, whereas the costs of those of SC1, SC2, and SC3 were about 0.031, 0.023, and 0.024 \$

Hence, it is clear SC4 shows a good thermal performance with a very low cost for cooking meals.

CRedit authorship contribution statement

M.A. Tawfik: Writing – review & editing, Writing – original draft, Supervision, Resources, Methodology, Investigation, Data curation, Conceptualization. **Atul A. Sagade:** Writing – review & editing, Writing – original draft, Validation, Methodology, Formal analysis, Conceptualization. **A.A. El-Sebail:** Writing – review & editing, Writing – original draft, Validation, Software, Formal analysis, Data curation. **A.M. Khallaf:** Writing – review & editing, Writing – original draft, Software, Methodology, Formal analysis. **Abhishek Saxena:** Writing – review & editing, Writing – original draft, Resources, Methodology, Data curation. **Hanan M. El-Shal:** Writing – review & editing, Writing – original draft, Validation, Investigation, Data curation. **W.E. Abd Allah:** Writing – review & editing, Visualization, Validation, Resources, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

Authors, M. A. Tawfik, Hanan M. El-Shal, and W.E. Abd Allah would like to thank and appreciate the administrative council of the Faculty of Agriculture, Zagazig University, Egypt, for offering all facilities to make this work possible. Atul A. Sagade acknowledge the research funding from university of Tarapacá through the Proyecto UTMAYOR N°5858-23.

Appendix A

$$M_1 = m_w C_w, \quad (A1)$$

$$a_1 = h_{cvw} n A_{vw}, \quad (A2)$$

$$\overline{f_1(t)} = h_{cvw} n A_{vw} T_v, \quad (A3)$$

Appendix B

$$M_2 = \frac{M_{equ}}{A_p}, \quad (B1)$$

$$a_2 = \frac{K_p}{X_p} (A_p + A_b), \quad (B2)$$

$$\overline{f_2(t)} = \left(\frac{K_p}{X_p} \right) (A_p T_p + A_b T_b), \quad (B3)$$

Appendix C

$$M_3 = m_w C_w, \quad (C1)$$

$$a_3 = h_{cvw} n A_{vw}, \quad (C2)$$

$$\overline{f_3(t)} = h_{cvw} n A_{vw} T_v, \quad (C3)$$

Appendix D

$$M_4 = \frac{M_{equ}}{A_p}, \quad (D1)$$

$$a_4 = -\frac{K_{pc}}{X_{pcm}} (A_p + A_b), \quad (D2)$$

$$\overline{f_4(t)} = -\left[\left(\frac{K_{pcm}}{X_{pcm}} \right) A_b T_b + \left(\frac{K_{pcm}}{X_{pcm}} \right) A_p T_p \right] \quad (D3)$$

Appendix E

For SC1 and SC2

$$R1 = \frac{1}{U_b} = \frac{X_p}{K_p} \quad (E1)$$

For SC3 and SC4. (Charging mode)

$$R1 = \frac{X_b}{K_b} \quad (E2)$$

$$R2 = \frac{X_b}{K_b} \quad (E3)$$

$$R3 = \frac{X_p}{K_p} \quad (E4)$$

(Discharging mode)

$$R1 = \frac{X_b}{K_b} \quad (E5)$$

$$R2 = \frac{X_b}{K_b} \quad (E6)$$

$$R3 = \frac{X_{PCM}}{K_{PCM}} \quad (E7)$$

For the four categories (SC1, SC2, SC3 and SC4)

$$R4 = \frac{1}{h_{cpv}} \quad (E8)$$

$$R5 = \frac{1}{h_{cvg}} \quad (E9)$$

$$R6 = \frac{1}{h_{cga}} \quad (E10)$$

$$R7 = \frac{1}{h_{cpa}} \quad (E11)$$

$$R8 = \frac{1}{h_{cva}} \quad (E12)$$

$$R9 = \frac{1}{h_{cpg}} \quad (E13)$$

$$R10 = \frac{1}{h_{cvw}} \quad (E14)$$

$$R11 = \frac{1}{h_{rgs}} \quad (E15)$$

$$R12 = \frac{1}{h_{cga}} \quad (E16)$$

Nomenclature

A_{air}	Surface area of the enclosed air (m ²)	T_{f2}	Fitted final temperature of the test load (°C)
A_b	Area of the back plate (m ²)	T_p	Temperature of the absorber plate (°C)
A_{bottom}	Area of the bottom of the cooking vessel (m ²)	T_s	Temperature of the sky (°C)
A_c	The aperture area (m ²)	T_v	Temperature of the cooking vessel (°C)
A_g	Area of the top inclined-glass cover (m ²)	T_w	Temperature of the cooking fluid (°C)
A_{fv}	Area between the cooking fluid and the cooking vessel (m ²)	$T_{w,max}$	Maximum temperature of the cooking fluid (°C)
A_{lv}	Lateral area of the cooking vessel (m ²)	U_b	Bottom loss coefficient (W/m ² K)
A_t	Area of the lid of the cooking vessel (m ²)	U_s	Side loss coefficient (W/m ² K)
A_p	Area of the absorber plate (m ²)	V	Wind speed (m/s)
CT	Cooking time available in solar noon (min)	X_p	Thickness of the absorber plate (m)
C_w	Specific heat of the cooking fluid (J/kg K)	X_{pcm}	Thickness of the phase change material (m)
Ex_i	Input exergy of SC (W)	Greek symbols	
Ex_o	Output exergy of SC (W)	α_a	Absorptivity of the enclosed air
Gr	Grasshof Number (–)	α_l	Absorptivity of the lid of the cooking vessel
h_{cag}	Convective heat transfer coefficient from the enclosed air to the glass cover (W/m ² K)	α_p	Absorptivity of the absorber plate
h_{cga}	Convective heat transfer coefficient from the glass cover to the ambient air (W/m ² K)	α_v	Absorptivity of the cooking vessel
h_{cva}	Convective heat transfer coefficient from the cooking vessel to the enclosed air (W/m ² K)	Δt	Time interval (seconds) (unless otherwise specified)
h_{cvw}	Convective heat transfer coefficient from the cooking vessel to the cooking fluid (W/m ² K)	ΔT	Temperature difference (°C)
h_{rgs}	Radiative heat transfer coefficient from the glass cover to sky (W/m ² K)	σ	Stefan-Boltzmann's constant ($\sigma = 5.6669 \times 10^{-8}$ W/m ² K ⁴)
I	Solar radiation (W/m ²)	τ_c	Characteristic boiling times (min)
$\bar{I}$	instantaneous solar insolation recorded over a time interval of Δt (W/m ²)	T_{f1}	Fitted initial temperature of the test load (°C)
I_{av}	Average solar radiation (W/m ²)	τ_s	Specific boiling times (min)
I_{ave}	A reference value of the average solar intensity equal 900 W/m ²	η_d	Daily efficiency (%)
$\bar{I}_s$	Average solar insolation over the cooking period (W/m ²)	η_{opt}	Optical efficiency of the concentrator (%)
I_T	The total solar radiation in the direct beam direction (W/m ²)	η_u	Utilization efficiency (%)
k_{air}	Thermal conductivity of the enclosed air (W/m K)	η_x	instantaneous exergy efficiency (%)
k_w	Thermal conductivity of the cooking fluid (W/m K)	$\eta_{x,daily}$	Daily exergy efficiency (%)
k_p	Thermal conductivity of the absorber plate (W/m K)	Abbreviations	
k_{pcm}	Thermal conductivity of the phase change material (W/m K)	LTCM	Lifetime cost of cooking a meal
M_w	Mass of the cooking fluid (kg)	PCM	Phase change material
N	Number of cooking vessels	SBC	Solar box-type cooker
n_s	Number of clear sun days (day)	SC	Solar cooker
P	Cooking power (W)	SC1	Solar cooker without TBPR and PCM

(continued on next page)

(continued)

Pr	Prandtl number (–)	SC2	Solar cooker with TBPR + without PCM
P_s	Specific cooking power (W)	SC3	Solar cooker without TBPR+ with PCM
Q_r	Amount of heat reflected from the TBPR (J)	SC4	Solar cooker with TBPR+ with PCM
T_a	Ambient air temperature (°C)	TBPR	Tracking-type bottom parabolic reflector
$\bar{T}_a$	Ambient air temperature during certain time (°C)	TES	Thermal energy storage
T_{air}	Temperature of the enclosed air (°C)	TPP	Thermal performance parameter
T_b	Back plate temperature (°C)		

Data availability

Data will be made available on request.

References

- [1] Omara AA, Abuelnuor AA, Mohammed HA, Habibi D, Younis O. Improving solar cooker performance using phase change materials: a comprehensive review. *Sol Energy* 2020;207:539–63. <https://doi.org/10.1016/j.solener.2020.07.015>.
- [2] Sagade NA, Sagade AA, Tawfik MA, Saxena A. Ensuring self-sustainability in decentralized communities through solar cookers combined with food dryers. *Sol Energy* 2023;260:83–93. <https://doi.org/10.1016/j.solener.2023.06.003>.
- [3] Tawfik MA, Oweda KM, Abd El-Wahab MK, AbdAllah WE. A new mode of a natural convection solar greenhouse dryer for domestic usage: performance assessment for grape drying. *Agriculture* 2023;13(5):1046. <https://doi.org/10.3390/agriculture13051046>.
- [4] Sagade AA, Sagade NA, Belgasim B, Tawfik MA, Kulkarni S, Shukla SK. Building integrated innovative non-imaging concentrating solar collector for clean cooking under low sun elevation. *Sustain Energy Technol Assess* 2023;57:103234. <https://doi.org/10.1016/j.seta.2023.103234>.
- [5] Tawfik MA, El-Tohamy M, Metwally AA, Khallaf AM, AbdAllah WE. Experimental and numerical investigation of thermal performance of a new design solar parabolic dish desalination system. *Appl Therm Eng* 2022;214:118827. <https://doi.org/10.1016/j.applthermaleng.2022.118827>.
- [6] Saxena A, Sagade AA, Joshi SK, Tyagi VV, Singh DB, Tawfik MA, Kumar A, Gupta P. A thermodynamic review on concentrating type solar cookers. *Sol Energy* 2025;286:113159. <https://doi.org/10.1016/j.solener.2024.113159>.
- [7] Singh M, Sethi VP. On the design, modelling and analysis of multi-shelf inclined solar cooker-cum-dryer. *Sol Energy* 2018;162:620–36. <https://doi.org/10.1016/j.solener.2018.01.045>.
- [8] Thakur A, Kumar R, Dwivedi A, Goel V. Solar cooking technology in India: identification and prioritization of potential challenges. *Renew Energy* 2023;219:119437. <https://doi.org/10.1016/j.renene.2023.119437>.
- [9] Cuce E, Cuce PM. A comprehensive review on solar cookers. *Appl Energy* 2013;102:1399–412. <https://doi.org/10.1016/j.apenergy.2012.09.002>.
- [10] Bondareva NS, Sheremet MA. Effect of nano-sized heat transfer enhancers on PCM-based heat sink performance at various heat loads. *Nanomater* 2020;10:17. <https://doi.org/10.3390/nano10010017>.
- [11] Keith A, Brown NJ, Zhou JL. The feasibility of a collapsible parabolic solar cooker incorporating phase change materials. *Renew Energy Focus* 2019;30:58–70. <https://doi.org/10.1016/j.ref.2019.03.005>.
- [12] Luo X, Guo Q, Tao Z, Liang Y, Liu Z. Modified phase change materials used for thermal management of a novel solar thermoelectric generator. *Energy Convers Manag* 2020;208:112459. <https://doi.org/10.1016/j.enconman.2019.112459>.
- [13] Mwaura MM, Thoruwa TFN. Phase change materials for energy storage in solar box cooker: sustainable innovations in energy technology. In: Proceedings of the sustainable research and innovation conference. Online; 2022. p. 214–9. <https://sri.jkuat.ac.ke/jkuatsri/index.php/sri/article/view/371/334>. [Accessed 16 March 2025].
- [14] Getnet MY, Gunjo DG, Sinha DK. Experimental investigation of thermal storage integrated indirect solar cooker with and without reflectors. *Results Eng* 2023;18:101022. <https://doi.org/10.1016/j.rineng.2023.101022>.
- [15] Coccia G, Aquilanti A, Tomassetti S, Comodi G, Di Nicola G. Design, realization, and tests of a portable solar box cooker coupled with an erythritol-based PCM thermal energy storage. *Sol Energy* 2020;201:530–40. <https://doi.org/10.1016/j.solener.2020.03.031>.
- [16] Mawire A, Lentswe K, Owusu P, Shobo A, Darkwa J, Calautit J, Worall M. Performance comparison of two solar cooking storage pots combined with wonderbag slow cookers for off-sunshine cooking. *Sol Energy* 2020;208:1166–80. <https://doi.org/10.1016/j.solener.2020.08.053>.
- [17] Coccia G, Aquilanti A, Tomassetti S, Muciaccia PF, Nicola G. Experimental analysis of nucleation triggering in a thermal energy storage based on xylitol used in a portable solar box cooker. *Energies* 2021;14(18):5981. <https://doi.org/10.3390/en14185981>.
- [18] Ose M, Staveland O, McGowan S, Unger JB, Christler NR, Weeman M, Schwart P. Phase change thermal storage: cooking with more power and versatility. *Sol Energy* 2021;220:1065–73. <https://doi.org/10.1016/j.solener.2021.03.040>.
- [19] Cuce PM, Kolayli S, Cuce E. Enhanced performance figures of solar cookers through latent heat storage and low-cost booster reflectors. *Int J Low Carbon Technol* 2020;15(3):427–33. <https://doi.org/10.1093/ijlct/ctz079>.
- [20] Palanikumar G, Shanmugan S, Chithambaram V, Gorjian S, Pruncu CI, Essa FA, Selvaraju P. Thermal investigation of a solar box-type cooker with nanocomposite phase change materials using flexible thermography. *Renew Energy* 2021;178:260–82. <https://doi.org/10.1016/j.renene.2021.06.022>.
- [21] Kumar A, Saxena A, Pandey SD, Joshi SK. Design and performance characteristics of a solar box cooker with phase change material: a feasibility study for Uttarakhand region, India. *Appl Therm Eng* 2022;208:118196. <https://doi.org/10.1016/j.applthermaleng.2022.118196>.
- [22] Alva G, Liu L, Huang X, Fang G. Thermal energy storage materials and systems for solar energy applications. *Renew Sust Energy Rev* 2017;68:693–706. <https://doi.org/10.1016/j.rser.2016.10.021>.
- [23] Chaichan MT, Kazem HA, Al-Waeli AH, Sopian K. Controlling the melting and solidification points temperature of PCMs on the performance and economic return of the water-cooled photovoltaic thermal system. *Sol Energy* 2021;224:1344–57. <https://doi.org/10.1016/j.solener.2021.07.003>.
- [24] Xu H, Romagnoli A, Sze JY, Py X. Application of material assessment methodology in latent heat thermal energy storage for waste heat recovery. *Appl Energy* 2017;187:281–90. <https://doi.org/10.1016/j.apenergy.2016.11.070>.
- [25] Tawfik MA, Sagade AA, Palma-Behnke R, El-Shal HM, AbdAllah WE. Solar cooker with tracking-type bottom reflector: an experimental thermal performance evaluation of a new design. *Sol Energy* 2021;220:295–315. <https://doi.org/10.1016/j.solener.2021.03.063>.
- [26] Tawfik MA, Sagade AA, Palma-Behnke R, AbdAllah WE, El-Shal HM. Performance evaluation of solar cooker with tracking-type bottom reflector retrofitted with a novel design of thermal storage incorporated absorber plate. *J Energy Storage* 2022;51:104432. <https://doi.org/10.1016/j.est.2022.104432>.
- [27] Mullick SC, Kandpal TC, Saxena AK. Thermal test procedure for box-type solar cookers. *Sol Energy* 1987;39:353–60. [https://doi.org/10.1016/S0038-092X\(87\)80021-X](https://doi.org/10.1016/S0038-092X(87)80021-X).
- [28] BIS. Bureau of Indian standards. In: Indian standards for solar cookers: IS13429; 2010. PART 1,2,3; 2010.
- [29] Collares-Pereira M, Cavaco A, Tavares A. Figures of merit and their relevance in the context of a standard testing and performance comparison methods for solar box – cookers. *Sol Energy* 2018;166:21–7. <https://doi.org/10.1016/j.solener.2018.03.040>.
- [30] Funk PA. Evaluating the international standard procedure for testing solar cookers and reporting performance. *Sol Energy* 2000;68(1):1–7. [https://doi.org/10.1016/S0038-092X\(99\)00059-6](https://doi.org/10.1016/S0038-092X(99)00059-6).
- [31] ASAE. Standard 580.1, testing and reporting solar cooker performance. 2013.
- [32] Panja PS, Samdarshi SK, Sagade AA, Jamal MR. Inclusivity of thermal performance parameters for evaluation of solar cookers and the necessity of uniformity in the appraisal using generalized test procedure for different designs. *Sustain Energy Technol Assess* 2024;68:103847. <https://doi.org/10.1016/j.seta.2024.103847>.
- [33] Tawfik MA, Sagade AA, El-Sebaai AA, Khallaf AM, El-Shal HM, AbdAllah WE. Enabling sustainability in the decentralized energy sector through a solar cooker augmented with a bottom parabolic reflector: performance modelling and 4E analyses. *Energy* 2024;287:129692. <https://doi.org/10.1016/j.energy.2023.129692>.
- [34] Wu SY, Xiao L, Cao Y, Li YR. A parabolic dish/AMTEC solar thermal power system and its performance evaluation. *Appl Energy* 2010;87:452–62. <https://doi.org/10.1016/j.apenergy.2009.08.041>.
- [35] Holman JP. Heat transfer. fifth ed. New York: McGraw-Hill; 1981.
- [36] Le Feuvre EJ, Ede AJ. Proceedings of the 9th congress. *Applied Mechanics* 1956. Brussels, Paper 1167.
- [37] Tiwari GN, Tiwari Arvind, Shyam. Handbook of solar energy: theory, analysis and applications. Singapore: Springer; 2016.
- [38] Duffie JA, Beckman WA. Solar engineering of thermal processes. fourth ed. New York: John Wiley & Sons; 2013.
- [39] Khalifa AMA, Taha MMA, Akyurt M. Solar cookers for outdoors and indoors. *Solar Energy* 1985;10(7):819–29. [https://doi.org/10.1016/0360-5442\(85\)90115-X](https://doi.org/10.1016/0360-5442(85)90115-X).
- [40] Olwi IA, Khalifa AMA. Numerical modeling and experimental testing of a solar grill. *J Sol Energy Eng* 1993;115(2):5–10. <https://doi.org/10.1115/1.2930025>.
- [41] Kumar N, Vishwanath G, Gupta A. An exergy based test protocol for truncated pyramid type solar box cooker. *Energy* 2011;36(9):5710–5. <https://doi.org/10.1016/j.energy.2011.06.046>.
- [42] Petela R. Exergy analysis of the solar cylindrical-parabolic cooker. *Sol Energy* 2005;79(3):221–33. <https://doi.org/10.1016/j.solener.2004.12.001>.
- [43] Sagade AA, Mawire A, Belgasim B, Tawfik MA, Sagade NA. Experimental performance evaluation of a parabolic dish solar geyser using a generalized approach for decentralized applications. *Sustain Energy Technol Assess* 2021;47:101454. <https://doi.org/10.1016/j.seta.2021.101454>.

- [44] Petela R. Exergy of undiluted thermal radiation. *Sol Energy* 2003;74(6):469–88. [https://doi.org/10.1016/S0038-092X\(03\)00226-3](https://doi.org/10.1016/S0038-092X(03)00226-3).
- [45] Pandey AK, Tyagi VV, Park SR, Tyagi SK. Comparative experimental study of solar cookers using exergy analysis. *J Therm Anal Calorim* 2012;109:425–31. <https://doi.org/10.1007/s10973-011-1501-1>.
- [46] Khallaf AM, Tawfik MA, El-Sebaai AA, Sagade AA. Mathematical modeling and experimental validation of the thermal performance of a novel design solar cooker. *Sol Energy* 2020;207:40–50. <https://doi.org/10.1016/j.solener.2020.06.069>.
- [47] El-Sebaai AA, Ibrahim A. Experimental testing of a box-type solar cooker using the standard procedure of cooking power. *Renew Energy* 2005;30:1861–71. <https://doi.org/10.1016/j.renene.2005.01.007>.
- [48] Sharshir SW, Elsheikh AH, Peng G, Yang N, El-Samadony MOA, Kabeel AE. Thermal performance and exergy analysis of solar stills—A review. *Renew Sustain Energy Rev* 2017;73:521–44. <https://doi.org/10.1016/j.rser.2017.01.156>.
- [49] Panwar NL, Kothari S, Kaushik SC. Techno-economic evaluation of masonry type animal feed solar cooker in rural areas of an Indian state Rajasthan. *Energy Policy* 2013;52:583–6. <https://doi.org/10.1016/j.enpol.2012.10.020>.
- [50] Indora S, Kandpal TC. Institutional and community solar cooking in India using SK-23 and scheffler solar cookers: a financial appraisal. *Renew Energy* 2018;120: 501–11. <https://doi.org/10.1016/j.renene.2018.01.004>.
- [51] Thollander P, Karlsson M, Rohdin P, Wollin J, Rosenqvist J. Investments, nonenergy benefits, and conservation. In: Thollander P, Karlsson M, Rohdin P, Wollin J, editors. *Introduction to industrial energy efficiency*. Academic Press, Elsevier Inc.; 2020. p. 147–58. <https://doi.org/10.1016/B978-0-12-817247-6.00007-9>.